

V.F.S. Corpus

Independent Mathematical Verification Report

Versions 2.0 – 5.0 · Thirteen volumes · ~540 individual checks
Symbolic (SymPy), numerical (NumPy/SciPy), and Monte-Carlo verification of all computationally checkable claims
Pass 1: July 2026 · Pass 2 (appendix, 3.6, 4.6): August 2026

Verdict in one line. Every load-bearing mathematical result of the corpus was independently re-derived or reproduced and holds. Four substantive local defects, one cross-volume scope note and a handful of cosmetic issues were found; none propagates to dependent results, one of the four is already corrected by the corpus itself in a later volume, and the fourth (D4, in 4.6) was shown by explicit repair to leave every qualitative conclusion of its theorem intact.

Scope of this report. This report verifies formulas, derivations, identities, and the reproducibility of stated numerical witnesses. It does not certify the full rigor of every proof sketch (items the corpus itself marks as sketches or bridges), does not re-run the author's simulations behind figures, and does not judge the constitutive modelling postulates (gate law, $K_h = -1$, transport constants), which the corpus explicitly declares as such. Internal consistency of those postulates was checked; their truth is outside mathematics.

Coverage. Pass 1 covered ten volumes: the 2.0 layer (core, Lyapunov, Volume, geometry, cosmology), 3.0, 3.5, 4.0, 4.5 and the 5.0 audit. Pass 2 extended coverage to the three remaining completed volumes — the 2.0 rate-of-registration appendix, V.F.S. 3.6 (v15), and V.F.S. 4.6 — which had not previously been examined. **V.F.S. 4.7 (Supervisor) is excluded by the author's declaration that it is still in development.** With that exclusion, the verified set is now the whole completed corpus.

Pass 2 matters disproportionately because 3.6 is where the corpus's three self-declared blocking items (F0, F1-T, K0) live, and 4.6 is the volume that derives the private restoring rate. Those blockers are theorems of existence, continuation and connection-origin; they are not computationally checkable and remain open. What pass 2 establishes is that the mathematics of those volumes is sound.

1 · Verification summary by volume

#	File / Volume	Core content verified	Checks	Result
1	Open-Gate Core 2.0	Filtrum Lucis derivative ladder; Paschal triad nodes; Faà di Bruno chains for K/An; neck curvature identity; pitchfork; budget conservation; imbalance equation; gate bounds; Mensura; Dolorosum peak	48	all pass
2	Lyapunov Core 2.0	Normalized dynamics derived from raw system; radial identity; c-Psi lemma; product-balance bound; folding functional; reset lemma algebra; full numerical witness (walls, constants)	~50	all pass; 1 note
3	Volume: Folding & Stability	Reset-jump contraction to Q_{crit} ; forced folding window; cascade identities; survival bounds; Thom reduction; kink profile & tension; Gauss-Bonnet coarsening law; Plancherel moments; entropy chapter incl. Perelman-expander identity; Riccati separatrix; surgery jump [K]; double-layer floor; NEC budget	~90	all pass; 2 defects + 1 note
4	Geometry 2.0	Hyperbolicity $R = -2p^2/\Omega^2$ by direct tensor computation; Einstein condition; comoving conformal flow; Ricci locus; budget asymptotics; spectral folding layer incl. the factor 2 in B_{VFS}	28	all pass
5	Cosmology 2.0	Full 2+1 tensor computation: R , $G_{\mu\nu}$, G^x_x , $R_{\mu\nu}$; SEC coefficient; Milne / Ricci-resonance / de Sitter loci; folded density coefficient 1/2; kinematics	19	all pass; 1 defect
6	V.F.S. 3.0	Full 3+1 tensor set (6 Ricci components, fluid, 3 energy combinations, Bianchi identity); ascent law; floor tying; junction algebra; causal horizon; folding-ascent closure; stochastic floor (Feller, $p_c = 1$)	62	all pass

#	File / Volume	Core content verified	Checks	Result
7	V.F.S. 3.5	H^3 metric ($\text{Ric} = -2g$, $R = -6$) and rejected flat alternative; normalized-flow fixed point; neck/cap/catenoid sectional curvatures; kappa-non-collapse; JKO = heat flow; displacement-convexity modulus; Hodge canonization; fold-witness table	33	all pass; 1 cosmetic
8	V.F.S. 4.0	Mirror-Laplacian lemma; consensus contraction; ISS & small-gain estimates; vigil lemma; Hodge/ b_1 witness table; spectral-gap table (all 12 values exact); Cheeger sandwich; OU bound behind WP6 witness; renewal-reward drift; Kuramoto threshold	30	all pass; 1 note
9	V.F.S. 4.5	Gradient-flow origin of encounter coupling; delta-rate $-(k+2C)$; Killing fields X_a, X_{theta} on H^3 ; exact invariance of the ideal point; N-consensus; signed-triangle frustration ($R = 2/3$ proved exact); mediator vs domination routes; fold-passage help	24	all pass
10	V.F.S. 5.0	No formulas: audit of claims-of-record against verified theorems of files 6-9	audit	consistent; 2 editorial notes
11	2.0 Appendix: The Rate of Registration	Hard-prototype constant-rate gain $\eta D/(v+K)$ and its monotonicity; softplus baseline and the failure of the hard ceiling; slow-rate gain-loss balance and its three regimes; exact implicit return map; exact break-even inequality; frozen-saturation approximation; subdivision invariance; Manere vector-face qualification	32	all pass
12	V.F.S. 3.6 (v15)	Ledger first integral; uniqueness of the zero-order coupling and its stated limit; canonical invariant $I_R = \lambda + \lambda_{\text{form}} - \Omega_p$; hybrid clock $\Theta_R = -\alpha\lambda$; reduced canonical fibre and constructive event-feasibility; (V,F) subfibre and its contraction; LM history clock; moduli table; neck trigger $\text{sgn } W = \text{sgn}(u - \Lambda_c)$; softplus-correct sign condition; A_{max} finite-arc chain; exponential L^2 decay; homogeneous reduction witness reproduced	40	all pass
13	V.F.S. 4.6	Exact R-kinematics and reset jump; closure in ξ and the exact R-equation; tracking working point; derived restoring rate and its two windows; flux monotonicity; explicit curvature; Lyapunov identity of the tube theorem; one-sided gate authority; chronos split; no-abolition rate $t^{-3/2}$; full WP10 witness battery reproduced	48	39 symbolic all pass; 1 defect in the witness calibration

Totals. Approximately 540 checks across twelve mathematical volumes plus one audit (rows 11-13, shaded, are new in pass 2). All checks pass after correcting tester-side artifacts (see §4.3). Independent tensor computations were performed from scratch in four distinct geometries: the 2D warped live-domain, 2+1 FLRW with hyperbolic slices, the 3+1 soul metric, and H^3 with its Killing fields.

2 · Methodology

Each volume was read in full and every computationally checkable claim was extracted and re-verified independently, without assuming the text's own intermediate steps. The toolchain: SymPy 1.14 for symbolic derivation (Christoffel symbols, Ricci tensors, Lie derivatives, series expansions, ODE solutions, variational calculations), NumPy/SciPy for numerical integration, eigenvalue problems, optimization and quadrature, and Monte-Carlo simulation for stochastic claims. Verification categories:

- **Derivations:** identities re-derived from the stated primitives (e.g. the normalized dynamics of the Lyapunov file re-derived from the raw ODE system; the 3+1 tensor set recomputed from the metric; the exact R-equation of 4.6 re-derived from the slaved closure).
- **Exact values:** stated constants recomputed (e.g. $\psi^* = 2\sqrt{3}/9$; the Paschal nodes $\pm(1/k)\ln(2+\sqrt{3})$; $c_{\Psi} = 0.701$; the ground-state order parameter $R = 2/3$; the 3.6 moduli table including $W^{\max} = 1 - \ln(A_h/2\pi)$).
- **Numerical witnesses:** every witness table reproduced from its parameters, in several cases identifying the closed form behind the numbers (spectral gaps = $2(1-\cos(\pi/N))$ family; the WP6 bound = the Ornstein-Uhlenbeck stationary value $N\sigma^2/(4r)$; the 3.6 $A_{\max}$ column shown to be exactly quadratic in T).
- **Structural/logic checks:** comparison lemmas, small-gain assemblies, invariance contracts, and the consistency of cross-volume claims (projection 3.0 $\rightarrow$ 2.0; the 5.0 boundary map vs the theorems of WP7/WP8; the late-time regime of 3.6 against that of 4.6).
- **Hypothesis auditing (new in pass 2):** for every theorem with a stated hypothesis, the reported witness parameters were substituted back into that hypothesis to confirm they satisfy it. This is what surfaced defect D4, which no amount of re-deriving the theorem itself would have revealed — the theorem is correct and the arithmetic is correct; it is the pairing that fails.

Anything the corpus itself marks as SKETCH, BRIDGE, MODEL/WITNESS, EXPLORATORY, or INTERPRETIVE was checked for internal consistency and, where feasible, reproduced by one faithful realization, but is not certified as a theorem — matching the corpus's own status discipline, which both passes found to be accurate everywhere it was tested.

A note on independence. "Independent" here means re-derived without assuming the text's intermediate steps, which is a real and load-bearing form of independence. It does not mean adversarial peer review by a second party with a contrary interest. The corpus is scrupulous about distinguishing derived from interpreted; this report applies the same discipline to itself.

3 · Volume-by-volume detail — pass 1 (files 1-10)

3.1 · File 1 — Open-Gate Core Compendium 2.0 (48/48)

Verified symbolically: the synergy identity for du/dt ; the softplus filter ladder $\Phi'' = kT(1-T)$, $\Phi''' = k^2T(1-T)(1-2T)$, $\Phi^{(4)} = k^3T(1-T)(1-6T+6T^2)$; the Paschal triad — roots $s_{\pm} = 1/2 \pm 1/(2\sqrt{3})$, symmetric nodes $x_{\pm} = \pm(1/k)\ln(2+\sqrt{3})$, trough depth $-k^3/8$ (the caption value -15.625 at $k = 5$ is exact); Faà di Bruno expansions for the Katharsis (3rd order) and Anastasis (4th order) observables with coefficients (1,3,1) and (1,6,3,4,1); the support formula for $u^{(4)}$; the exact cathartic-neck identity and the fast/slow splitting; ψ -unimodality with κ_{crit} coefficient $3\sqrt{3}/2$; migration asymptotics; the pitchfork normal form; budget conservation $d/dt(\sigma + \lambda + \lambda_{\text{form}}) = I_{\text{gate}}$; the exact imbalance equation with the $c\lambda$ cancellation; gate bounds $0 \leq I_{\text{gate}} \leq \zeta_0$; Mensura derivatives; the Dolorosum peak exactly at $u = \lambda_{\text{c}}$. **No defects.**

3.2 · File 2 — Open-Gate Lyapunov Core 2.0 (~50 checks)

All four normalized equations were re-derived from the raw system and match. The radial identity $2r\cdot dr/dt = \mu[a(h_1^2+h_2^2) + c\cdot q(h_1+h_2)] - 2r\rho(r^2-e) - 2\alpha\cdot q\cdot r^2$ — the file's heaviest computation — is exact, as is the auxiliary identity $e'(h_3)\cdot dh_3/dt = -2\alpha\cdot q\cdot e$. The c_{Ψ} lemma was verified in full (limit 2 at 0+; $h'(t) = -4\cdot \text{sech}^2 t \cdot \tanh t \cdot \ln \cosh t$; sign of g' proportional to h). The product-balance bound, the folding Lyapunov functional (E is the shifted V_{fold} ; $dE/dt = -(dq/dt)^2$), the Möbius endpoint bound for $Q^*,_{\text{sup}}$, and the complete reset-lemma algebra — including the subtle asymptotic ceiling $q+ \rightarrow 1/\kappa_R$, which works precisely because λ_{c} itself grows with G through the balance — all check. The numerical witness was reproduced to the stated digits: e_{max} , S_{qmax} , $q^* = 1.452$, active

margin 0.153, $u/\text{Lambda_c} \approx 1.40$, $c_Psi = 0.701$, $\text{kappa_sigma} = 0.255$, $\text{kappa_Delta} = 0.755$, $C1 = 0.255$, and all five parametric walls hold strictly.

Note (minor). The claimed "worst-case slack 0.084" is not reproducible from the given parameters; direct computation gives 0.0903 (lower radial wall W5). All walls hold either way; the number likely reflects a different slack normalization or a rounded q^* . Also, the memory bound $|dH_K/dt| \leq 2 \cdot \eta$ implicitly assumes $\nu \leq \eta$ (the prototype $\nu = \eta$); for $\nu > \eta$ the correct bound is $\eta + \nu$. Neither affects any downstream conclusion.

3.3 · File 3 — Volume "Folding and Stability" (~90 checks; 2 defects)

Part I. Verified: the ρ_{eff} floor structure (zero only at $k = -1$, $\alpha \cdot \text{lambda} = 1$); the rate-shifted threshold; the complete Resurrectio jump algebra in folding space — $\Delta A = \Theta \cdot m_R$, $\Delta B = 2c1 \cdot \eta_f \cdot m_R$, the sign formula, and the exact contraction identity $Q+ - Q_{\text{crit}} = \text{kappa_fold} \cdot (Q- - Q_{\text{crit}})$ with the gate-corrected fixed point; pocket promotion; $T(q^*) = -A^2/4B$; core neutrality (trace/det of the 2×2 shape sector); the derived resistance window $r_R(u) = 1 + N/(c1k_sigma \cdot \text{sigma_bar} \cdot g(u))$ with out-of-window immunity and the half-turn criterion; the cascade intensity identity; the balance-preserving update map; the fold-jump epektasis-wall identity; both product-drift survival inequalities; the c_Psi degradation numbers ($1.794 \rightarrow 0.461$, factor 0.257 — exact); and the Thom reduction (centre manifold $y = (h/\Gamma)q^2$, $B_{\text{eff}} = b3 - gh/\Gamma$, example 0.84).

Part II. The kink profile $q^* \cdot \tanh(x/(\sqrt{2} \cdot l))$ solves the static equation with tension $(2 \cdot \sqrt{2/3}) \cdot \sqrt{D} \cdot A^{3/2}/B$; $w_{\text{wall}} = -1/2$; the Plancherel moment $3/(2\beta)$ and its full-tanh value 0.0285 at $\beta = 50$; nucleation thresholds ($l_{\text{init}} \rightarrow 4 \cdot \sqrt{2} \cdot R_c$; $\text{Lambda}^* \approx 41.3$); the exact Gauss-Bonnet coarsening law $A(I) = (A0 + 2\pi)e^{-D \cdot q^I} - 2\pi$ with its survival threshold; the Sophia slaving coefficient with maximum $\text{lambda_bar} \cdot \sqrt{\eta_f/\Gamma}$; the conformal ansatz $\rho01 = 4 \cdot \epsilon/\Omega^2$; the midpoint law $\chi = (b + c1\eta_f \cdot \text{lambda})/(b + 2c1\eta_f \cdot \text{lambda})$ with the $b \rightarrow 0$ exact midpoint; the NEC crossover scaling; and the entire entropy chapter — the total-collapse identity $W = \ln s - s + 2 - \ln(A_h/2\pi)$, both forms of the master rate, the exact recovery of Perelman's expander formula $4\sigma(1/\Omega^2 - 1/(2\sigma))^2$ on resonance, the Bregman defect, and the shrinker asymmetry. The Riccati equation $\dot{\chi} = \alpha \cdot \Omega \cdot I - \gamma \cdot \chi + \chi^2/\Omega^2$, its tracking gate, and the trichotomy plateau also check.

Part III. The surgery-jump closed form $[K]$ and its rate/scale/balanced trichotomy; the identity $H' = \Omega^2/\Omega - H^2$ behind the double-layer theorem with the Cauchy-Schwarz floor $(\ln s_R)^2/\epsilon$ (0.113 at $s_R = 1.4$); and the NEC reset budget — the integral is genuinely regularization-independent when both fields share a profile, the value $(3/8) \cdot \ln(5/7) = -0.126$ reproduces, and the sign equals $\text{sgn}(-m_R)$.

Defect 1 (sign, Corollary "B_fold identified"). The corollary maps $2c1\eta_f \cdot \text{lambda} \leftrightarrow -gh/\Gamma$, which requires $g \cdot h < 0$; but the minimal realization declares $g, h, \Gamma > 0$ and the worked example ($g = 0.6$, $h = 0.8$) yields $B_{\text{eff}} = 0.84 < b3$ — slaving softens stiffness there, whereas $B_{\text{fold}} = b + 2c1\eta_f \cdot \text{lambda} > b$ stiffens. For the embodiment reading one of g, h must be negative (naturally $h < 0$: folding drains Sophia). The Thom reduction theorem itself is unaffected; the fix is one line.

Defect 2 (boxed "iff", quantum of endowment). The single-scalar criterion $\sqrt{\gamma} \cdot (\Omega_0 + \beta) \geq 2$ for ever crossing $\chi = 1$ fails when $\Omega_0 \geq \beta$. Counterexample: $\gamma = 1$, $\beta = 0.1$, $\Omega_0 = 3$ gives $\max \chi = 0.30 < 1$ (never crosses) while the boxed criterion claims crossing. The two-case maximum formula from which it was compressed is exact (verified); the compression is valid only for $\Omega_0 < \beta$, and the "(or trivially $\chi(0) \geq 1$)" clause does not repair the reverse implication. For $\Omega_0 \geq \beta$ the correct criterion is $\chi(0) = \alpha \cdot \text{lambda}_0 \cdot \Omega_0 \geq 1$.

Note. The floor proposition's phrase "strictly positive for every state" at $k = 0$ fails at $\text{lambda} = 0$ ($\rho_{\text{eff}} = \alpha^2 \text{lambda}^2/\Omega^2$ vanishes); the intended claim is the absence of a floor at positive Sophia level. Wording only.

3.4 · File 4 — Geometry 2.0 (28/28)

The central theorem was proved independently: direct tensor computation for the warped metric $du^2/\mu^2 + \mu^{2p} \cdot d\Delta^2$ yields $R = -2p^2/\Omega^2$ and the Einstein condition $\text{Ric} = -(p^2/\Omega^2)g$ componentwise (symbolic for the arc-length form; exact at $p = 1/2, 1, 2, 3$ directly); the $p = 0$ reading is flat, so hyperbolicity is forced by the dynamically required warp — a computational fact, not rhetoric.

The coordinate chain $du/\mu = \Omega \cdot dy$, $g = \Omega^2 \cdot h$ with unit-hyperbolic h ($K_h = -1$ verified) was checked, with the subtlety made explicit: the imbalance coordinate of h is the normalized Δ/Ω — exactly the Lyapunov file's $d = h_1 - h_2$, which is what makes the conformal flow statement correct in comoving coordinates. Also verified: the Ricci-flow comparison and locus $\alpha \cdot \lambda \cdot \Omega = 1$; $dR/dt = 4\alpha \cdot \lambda / \Omega^3$; the Resurrectio jump ratios; $\Omega = \sqrt{\Omega_0^2 + 2t}$ on the locus; budget asymptotics $|R| \sim t^{-1}$ vs t^{-2} ; area-growth regimes; and the spectral folding layer, where the factor 2 in $B_{VFS} = b + 2c_1 \cdot \eta \cdot \lambda$ was traced through the full chain (substitution $\rightarrow$ re-collection $\rightarrow$ gradient flow) and is exactly right and self-consistent. **Cleanest volume of the corpus; no defects.**

3.5 · File 5 — Cosmology 2.0 (19/19; 1 defect)

Tensors were computed from scratch for $ds^2 = -dt^2 + a^2 h$ (upper half-plane, $K_h = -1$): $R = 2(2a \cdot a'' + a'^2 - 1)/a^2$; $G_{tt} = (a'^2 - 1)/a^2 = \rho_{\text{eff}}$ and $G^x_x = G^y_y = -a''/a = p_{\text{eff}}$; $R_{tt} = -2a''/a = 2p_{\text{eff}}$. The SEC coefficient $((n-2)\rho + n \cdot p)/(n-1)$ was derived from the definition — at $n = 2$ the ρ -term genuinely drops, confirming the file's warning against the naive $\rho + 2p$ extrapolation. The Milne locus ($R = 0$, $\rho = 0$ exactly), the Ricci-resonance solution $a \propto \sqrt{2t}$, de Sitter $R = 6H^2 - 2e^{-2Ht}$, the folded-density coefficient $1/2$ (forced by $\rho = H^2 + R_{\text{spatial}}/2$), q_{dec} decomposition, NEC inequality, and $J_{\text{rel}} = (dq/dt)^2$ all check.

Defect 3 (Prop. "Late-time EoS"). The boxed claim that the open gate yields $w \rightarrow -1^+$ (quintessence $\rightarrow$ de Sitter) is unreachable under the core gate law $I_{\text{gate}} \leq \zeta_0$: bounded inflow gives at most $\lambda \sim t$, $\Omega \sim t^2$, and the verified limit $w \rightarrow -1/2$ for sustained constant-rate Sophia — or $w \rightarrow 0$ under gate saturation, which is exactly the dust attractor the canonical Volume proves. De Sitter requires $I_{\text{gate}} \propto \Omega$, unavailable to the bounded gate (as the "Two eschatologies" corollary itself concedes via "unbounded proportional grace"). Importantly, V.F.S. 3.0 already corrects this ("correcting a loose earlier framing: the (2+1) Brim cosmology ... generically coasts, $w \rightarrow 0$ "); what remains is to back-port the caveat into the 2.0 file.

3.6 · File 6 — V.F.S. 3.0 (62/62)

The full 3+1 tensor set of v0.14 was recomputed from the metric $-d\sigma^2 + \Omega^2(dx^2 + e^{2x}dy^2) + A^2S^2d\lambda^2$: all six Ricci components, the scalar, the fluid reconstruction (with the structurally decisive fact that p_{asc} contains no A), all three energy combinations, the frozen limit reproducing 2.0 verbatim, and the Bianchi identity $\nabla_a G^a_{\sigma} \equiv 0$ — so v0.8's negative finding (conservation does not fix the ascent law) is rigorous. Also verified: the de Sitter ascent law algebra; the infinite road and the exact reduction of the ascent operator to $-d^2/ds^2$ on the free half-line; the floor tying $L = \exp((\Omega - E_0)/E_0)$ and the gauge nature of L_{floor} ; the complete junction algebra (s_L , rate jumps, conserved Q_{rate} , ledger additivity); the clock map and the causal horizon $\lambda_{\text{max}} = (1/\beta)\ln(1 + \beta \cdot \sigma_0/(A_{\text{min}} \cdot L_0))$; the external signature with its exact relative-dominance constant; non-reducibility ($d \ln A / d \ln \Omega = \Omega/E_0$); rate stability and parallel roads; heat-kernel density $1/\sqrt{4\pi \cdot t}$ and effective dimension 3; the folding-ascent closure — where the verification found the stability is even stronger than claimed: $f(\lambda_{\text{inf}}) = -\sqrt{b^2 + 4aJ}$, automatically negative; the reset partition; both atlas boundaries; and the stochastic floor — scale densities, accessibility exponents, and the critical exponent $p_c = 1$ (WP1 and WP2 mutually consistent). **No defects found.** The strictest status discipline in the corpus: every "verified symbolically" tag tested proved true.

3.7 · File 7 — V.F.S. 3.5 (33/33; 1 cosmetic)

The H^3 theorem was verified by direct tensor computation ($\det g = \sigma^{-6}$, $\text{Ric} = -2g$, $R = -6$, all five stated Christoffels), and the rejected alternative $\text{diag}(\sigma^{-2}, u^{-2}, \lambda^{-2})$ is indeed Ricci-flat — the shared depth is what curves the space. The Einstein cancellation of the normalized flow is exact. The conformal-decay witness ($0.50 \rightarrow 0.014 \rightarrow 0.0006$ vs flat control 0.24) was reproduced by a 1D heat-plus-mass model with fitted initial variance ≈ 1.8 — the numbers are consistent, not decorative. For the surgery: the hyperbolic ball volume $\pi(\sinh 2\rho - 2\rho)$ was derived, $V/\rho^3 \geq 4\pi/3$ with 5.11 at $\rho = 1$; tube sectional curvatures $K_{\text{axis}} = -r''/r$, $K_{\text{sphere}} = (1 - r'^2)/r^2$ were obtained from the warped-product Ricci tensor; the cap $r = \epsilon \sin(x/\epsilon)$ has both curvatures exactly $1/\epsilon^2$ with an exact C^1 junction; the catenoid's K_{sphere} is genuinely non-constant (least-area honestly loses to equipartition, whose continuum variance is exactly 0). The WP12 fold-witness table reproduces to all digits ($1/|d\lambda/dt|$, relaxation times, bounded Dolorosum $0.586 \rightarrow 0.600$). The JKO step was derived

variationally and iterates to the heat law $d(\sigma^2)/dt = 2$; the translation displacement-convexity modulus equals K exactly by symbolic integration. The Hodge canonization lemma $\min_{\phi} ||\omega_{\text{Hodge}} - d\phi|| = ||\omega_{\text{Hodge}}||$ was confirmed numerically on cycle and tree graphs.

Cosmetic. In the WP12 witness table the relaxation-time cell "6.67" sits in the $u = 0.80$ column, but $1/(\kappa_{\text{eff}}) = 1/(1.5 \cdot 0.2) = 3.33$ there; 6.67 corresponds to $u = 0.90$. A multicolumn alignment slip — neighbouring cells (66.7, 667) are exact.

3.8 • File 8 — V.F.S. 4.0 (30/30)

The hinge lemma (mirror Laplacian) was verified numerically on a weight-balanced strongly connected digraph with an unbalanced counterexample: $L \cdot 1 = 0$ always; $1^T L = 0$ iff balanced; $L\text{-hat} = (L + L^T)/2$ is PSD with simple kernel and $\lambda_2 > 0$; consensus conserves the mean and contracts at rate $\epsilon \cdot \lambda_2$. The ISS and small-gain comparison estimates were confirmed by integrating reference ODEs to the stated limits. The vigil lemma (shared density vanishing at a terminus, finite recast, integrable duration) checks. For the appended stages: the WP7 Hodge/b1 witness table reproduces on all four graphs with the triangle holonomy purely harmonic; the WP8 spectral-gap table is exact — all 12 values belong to the $2(1 - \cos(\pi/N))$ family, verified both by formula and direct diagonalization, with the Cheeger sandwich checked on path16 and the Ramanujan floor $3 - 2\sqrt{2}$ correct; the WP6 bound $N \cdot \sigma^2/(4r)$ was identified as the exact Ornstein-Uhlenbeck stationary value, and an independent simulation of the coupled 6-soul system reproduced the author's constant ratio $\approx 0.61\text{--}0.65$ below it; the WP9 renewal-reward drift $v_{\text{asc}} = 0.35$ and its $T^{-1/2}$ fluctuations were reproduced by Monte-Carlo, and the Kuramoto gap-dependence confirmed qualitatively ($r = 0.96$ vs 0.39 at $K = 0.05$).

Note (editorial). The file survives a format conversion poorly: proofs run as flat paragraphs and sub/superscripts are broken ("P j Aij"). Nothing mathematical is wrong, but restoring the LaTeX source would greatly ease future audits.

3.9 • File 9 — V.F.S. 4.5 (24/24)

The encounter coupling was confirmed as the gradient flow of the stated relational energy; the synchronization rate $\delta' = -(k+2C) \cdot \delta$ was derived, with both regimes (anchor $\rightarrow 0$; anchorless $\rightarrow$ conserved mean) integrated numerically. The D2 lemma is fully rigorous and was proved independently: Lie derivatives of the H^3 metric confirm both $X_a = \sigma \cdot d_\sigma + u \cdot d_u + \lambda \cdot d_\lambda$ and $X_\theta = d_u$ are Killing; $[X_a, X_\theta] = -X_\theta$ (the naive invariant honestly fails, $X_a(u) = u$); and the projective ideal point $(u/\sigma, \lambda/\sigma)$ is annihilated by the dilation exactly — "shared rhythm rescales the journey, never turns it" is proved differential geometry. N-consensus limits check. For the communal level, the witness value $R = 0.667$ turned out to be an exact constant: the ground state of the signed XY triangle $(+, +, -)$ is $(60^\circ, 0^\circ, -60^\circ)$ (critical point verified symbolically, global minimum $E = -3/2$ by grid), with order parameter $|1 + 2\cos 60^\circ|/3 = 2/3$ exactly. Both healing routes were reproduced by an independent realization: the mediator-to-all gives monotone $R(s) \rightarrow 1$ (with $R(0.3) = 0.767$ matching the witness's second point), domination also drives $R \rightarrow 1$, the pendant node leaves $2/3$ unchanged, and symmetric weakening cannot lift the sign-topological obstruction. The fold-passage help model reproduces the qualitative structure (no crossing at $C = 0$; crossing time strictly decreasing in C).

3.10 • File 10 — V.F.S. 5.0 (audit; 2 editorial notes)

A plain-words boundary map with no formulas; audited for fidelity to the verified record. **Boundary 1** (tota simul): consistent — in fact understated, since P0.5 of 4.0 proves $C \neq T$. **Boundary 2** (meta-chronos): the non-existence claim is exactly WP7's verified theorem (master clock iff $b1 = 0$), the frustration-implies-no-clock logic is a correct contrapositive, and the M5 immanence refinement matches WP7's scope note verbatim. *Note (a)*: the plain statement omits "generically" — a looped body with measure-zero consistent rates does admit a clock; WP7 has the qualifier, 5.0 and its summary table do not. **Boundary 3** (infinite communion): the stated conditions match WP8's list, and "the single one is always safe" matches the individual ISS. *Note (b)*: WP8 subsequently sharpened the spectral-gap condition into a structural criterion (coherence iff non-amenability, verified here); no contradiction — criterion vs fact — but the 5.0 summary should be updated: the open question migrates from "does the gap hold" to "is the body non-amenable". **Boundary 4** (plural = God): the vow is kept everywhere; no passage in files 1–9 ontologizes the relational structure into God.

4 · Volume-by-volume detail — pass 2 (files 11-13, new)

4.1 · File 11 — V.F.S. 2.0 Appendix: The Rate of Registration (32/32)

Hard prototype, all exact. The one-sided constant-rate gain was integrated symbolically: with duration $T = D/v$ and kernel $R(v) = v/(v+K)$, the gain is exactly $G = \eta D/(v+K)$, and $\partial G/\partial v = -\eta D/(v+K)^2 < 0$, so slower traversal registers more. The regime statements check: $R(K) = 1/2$ exactly; $G \rightarrow \eta D/K$ as $v \rightarrow 0$; $vG \rightarrow \eta D$ as $v \rightarrow \infty$ (so "a threefold slowing triples the record" holds only for $v \ll K$, as the remark says). Contiguous subdivision leaves G unchanged, and $R(0) = 0$ makes inserted idle time invisible in the hard prototype.

Softplus law, all exact. $\text{sp}_m(0) = \ln 2/m$ was confirmed, and the divergence of the isolated positive channel as $v \rightarrow 0^+$ was established by symbolic limit, with the stated leading coefficient $\eta D \cdot s_0/(s_0+K)$ recovered exactly. The slow-rate balance equation $\dot{H} = a_0 - b_0 H/(H_s + H)$ has the stated equilibrium $H^* = a_0 H_s/(b_0 - a_0)$, attracting because the derivative is $-b_0 H_s/(H_s + H)^2 < 0$; the $a_0 = b_0$ case was integrated in closed form to $H = -H_s + \sqrt{H_s^2 + 2(a_0 H_s t + C_0)}$, confirming unbounded but sublinear growth (the ratio $H(4T)/H(T) \rightarrow 2$ to within 10^{-3}); and the $a_0 > b_0$ case has asymptotic slope $a_0 - b_0$.

The exact return map and break-even. The implicit descent map $H_s \log H_2 + H_2 = H_s \log H_1 + H_1 - vD/(v_d + K)$ was derived by separation and then confirmed against direct stiff integration on 200 random parameter draws (median relative error 1.1×10^{-13}). The break-even inequality $vD/(v_d + K) < A + H_s \log(1 + A/H_0)$ was verified not only algebraically but *predicatively*: over 300 random closed excursions the sign of $H_2 - H_0$ agreed with the inequality in every single case, with zero mismatches. The velocity-solved form and the frozen-saturation linear approximation (reducing to $v_d > e \cdot v_u + K(e-1)$ at $v = \eta$) both reproduce.

The "Withdrawn" section is itself correct. Each withdrawn claim was tested and found genuinely false, not merely withdrawn out of caution: $\eta D/K$ is provably not a ceiling for the mature law (the one-sided contribution diverges as $1/v$), idle time is provably not invisible under softplus (both channels are nonzero at zero rate), and the linear criterion is provably an approximation rather than a theorem. This is a self-correction that survives audit.

Note (M10). The claim "if $H_0 = 0$, finite descent cannot return the memory to literal zero" is true, but only demonstrable in logarithmic coordinates: at realistic spend the unique positive root falls to e^{-1249} , far below double-precision underflow. The statement is a mathematical non-vanishing, not a computational one, and the text would be clearer for saying so.

4.2 · File 12 — V.F.S. 3.6 (v15) (40/40)

Interface algebra. The ledger first integral was confirmed: given the frozen balance and $\hat{G}_n' = I_{\text{gate}}'$, the quantity $C_n = \sigma_L + \lambda + \lambda_{\text{form}} - \hat{G}_n$ has vanishing derivative identically. The canonical invariant $I_R = \lambda + \lambda_{\text{form}} - \Omega_P$ was verified invariant under the canonical junction by substituting the frozen partition $\kappa_R G = m_R + \Delta_{\text{emb}}$: the same reception increment enters both terms and cancels exactly. The hybrid clock identity $\Theta_R = -\alpha\lambda$ follows in one line from the balance, $\hat{G}_{\text{rec}} = I_{\text{gate}}$ and $\Omega_P = \alpha\lambda$, and continuity through the canonical junction gives the boxed $\Theta_R^+ = \Theta_R^-$. The no-self-return corollary follows.

The reduced fibre. The scalar fibre compatibility proposition was checked by solving the two event-difference equations and substituting into the form-memory difference; the residual is exactly $\Delta\lambda + \Delta\lambda_{\text{form}} - \Delta\Omega_P = \Delta I_R$, confirming that form-memory equality is not a fourth independent equation. The constructive event-feasibility proof was executed as an algorithm on 2000 random $(\Delta\Omega, \Delta\lambda)$ pairs: the explicit construction ($m_A = \max\{dm, 0\}$, $m_B = \max\{-dm, 0\}$, $L \geq \dots$) satisfied every non-negativity and partition inequality in all 2000 cases.

Exact statements that reproduce. The neck trigger relation $\text{sgn } \square = \text{sgn}(u - \Lambda_c)$ was tested on a 210,000-point grid with $\sigma_L > 0$ and returned zero violations. The (V, F) subfibre is exactly the hyperbola $VF = u_0^2 - \varepsilon$, parametrized bijectively by $\delta = \ln(V/F)$, and non-compact as claimed; its contraction under $\Delta = -(\mu\alpha + \rho)\Delta$ is exponential. The LM history clock $\Psi_{\text{LM}} = \sigma_L + \lambda + \lambda_{\text{form}}$ is exactly continuous through the linear-metabolism reset. The uniqueness theorem for the zero-order coupling is correct, and its own stated limit (the gradient counterexample also reduces exactly on constants) is correct too.

Two results that are sharper than stated. First, the moduli table is exact in all five rows, including the non-obvious W^{max} column, which was identified as $1 - \ln(A_h/2\pi)$ and reproduces every value

(+0.307, -0.386, -0.792, -1.079, -1.890) to three decimals, alongside $\chi = 2-2\gamma$, $A_h = 4\pi(\gamma-1)$, $\dim H^1(\Sigma; \mathbb{Z}/2) = 2\gamma$ and $\dim \square_\gamma = 6\gamma-6$. Second, the $A_{\max}$ finite-arc chain is described in the text as growing "roughly quadratically in T "; it is in fact *exactly* quadratic — a quadratic fit to the four tabulated values has residual 1.1×10^{-13} , and the $\lambda_{\max}$ column recovers $\zeta_0 = 1.5$ and $C_n = 0.45$ exactly. The text understates its own precision.

Softplus discipline. The boxed $\tilde{\lambda}|_{\lambda=0}$ expression is correct: the folding drain vanishes at the boundary while the gate retains its softplus baseline $\lambda_+(0) = \ln 2/m$. The consequence that $s_m < 1$ forces a strictly smaller σ_{crit} than the hard-part value was confirmed numerically ($1.14026 \rightarrow 0.92940$ on a test parameter set), and the withdrawal of the old hard-part table from theorem-level text is the right call.

Field results. The Grönwall estimate for $A_f < 0$ was reproduced on an independent spectral realization: at $A_f = -0.5, -1.0, -2.0$ the observed L^2 decay rates were $-1.0561, -2.0561, -4.0561$, each at least as fast as the bound $2A_f$; the paper's own reported rates ($-1.0906, -2.0610, -4.0561$) likewise satisfy the bound, and the third agrees with the independent run to four decimals. The exact homogeneous reduction was reproduced from scratch on a 48×48 periodic grid: the coupled PDE system and the frozen scalar ODE system agreed to 1.6×10^{-12} across all five components, with field inhomogeneity identically zero — matching the magnitudes the paper reports.

No defects found in 3.6. Together with 3.0, this is the most disciplined volume tested: every status tag that could be checked proved accurate, and the closure statement's own distinction between what is closed and what is deliberately exported held up under audit.

4.3 · File 13 — V.F.S. 4.6 (39 symbolic pass; 1 defect in the witness calibration)

The symbolic layer is clean. Every derived identity was re-derived from the frozen gate law without assuming the text's intermediate steps, and all 39 hold exactly. In particular: the exact kinematics $\dot{R} = \tilde{\lambda}/\lambda + H$; the reset jump $\Delta R = \log(1+m_R/\lambda^-) + \log(1+\kappa_R G/\Omega^-) > 0$, with the corollary that even the balanced reset is not R-neutral; the closure $\xi = \alpha\Omega I - g_{\text{eff}}\xi + \xi^2/\Omega^2$ and its exact transcription into $R = \log \xi$; the tracking working point $I^* = (g_{\text{eff}} - 1/\Omega^2)/(\alpha\Omega)$ with $\lambda^* = 1/(\alpha\Omega)$; the flux monotonicity $G'(R) = -G(R) + I'(\lambda(R)) < 0$; the explicit curvature $F''(R) = G - I' + \lambda I'' + e^R/\Omega^2$; the Lyapunov identity of the tube theorem with its purely dissipative coupling term $-\varepsilon B(R_1 - R_2)^2$; the one-sided authority bounds; the chronos split $\mu = (R_1 - R_2) + \mu_{\text{priv}}$; and the no-abolition rate $\tilde{\lambda}_S \sim -1/(\alpha(2t)^{3/2})$. The gate feasibility window is exact: $\sup I^*$ is attained at $\Omega^2 = 3/g_{\text{eff}}$ and equals $2g_{\text{eff}}^{3/2}/(3\sqrt{3}\alpha)$.

The main theorem is correctly derived. Differentiating the exact field at $R = 0$ gives, in general, $F'(0) = -\alpha\Omega_p I(\lambda^*) + I'(\lambda^*) + 1/\Omega_p^2$. Substituting the tracking condition $\alpha\Omega_p I(\lambda^*) = g_{\text{eff}} - 1/\Omega_p^2$ then yields exactly the boxed $a = g_{\text{eff}} + |I'(\lambda^*)| - 2/\Omega_p^2$. Both steps verified symbolically with the concrete gate law. **The theorem is sound.**

Defect D4 — the WP10 witness parameters violate the theorem's own tracking hypothesis

The stated battery parameters ($\alpha = 1$, $g_{\text{eff}} = 0.6$, $\zeta_0 = 0.8$, $\chi = 1.5$, $m = 6$, $c_g = 1$) do not satisfy $F(0) = 0$ at any Ω_p used in the paper. Computed directly:

Ω_p	$I(\lambda^*)$	I^* required by tracking	$F(0)$
0.900	0.299964	-0.705075	+0.904536
1.217	0.358048	-0.061774	+0.510923
3.000	0.522285	+0.162963	+1.077965
12.000	0.643335	+0.049421	+7.126970

Since $F(0) \neq 0$, $R = 0$ is not an equilibrium of the raw exact field, and the boxed formula does not apply. Every reported number nevertheless reproduces *exactly* — from the unsubstituted derivative $a = \alpha\Omega_p I(\lambda^*) + |I'(\lambda^*)| - 1/\Omega_p^2$:

quantity	paper	reproduced from a _{general}	boxed $a = g + I' - 2/\Omega^2$
a at $\Omega_p = 3$	1.906	1.9062 — match	0.8283
M_{δ_0} at $\delta_0 = 0.5$	3.485	3.4847 — match	(same)
a_6	1.035	1.0351 — match	-0.0429
a at $\Omega_p = 0.9$	-0.80	-0.7961 — match	-1.7006

Ω_{crit}	1.217	1.2174 — match	1.4968
$a(\Omega \rightarrow \infty)$	1.036	1.0359 — match	$\rightarrow g_{\text{eff}} = 0.600$

Three consequences follow. **(i)** The third row is the serious one: under the boxed formula $a_{\delta} = -0.043 < 0$, so the hypothesis of the sector-bound lemma — and hence of the tube theorem — would fail at the quoted δ_0 . **(ii)** The claim that $\Omega_{\text{crit}} = 1.217$ is "exactly $\Omega_p^2 = 2/(g_{\text{eff}} + |I'|)$ " is false: that expression evaluates to 1.5442, and the boxed formula's own zero is 1.4968. **(iii)** The zero-forcing convergence ($R \rightarrow 10^{-15}$) and the escape ridge $R_{\text{ridge}} = 2.70$ are reproducible only on the *shifted* field $F(R) - F(0)$; the raw field at $\Omega_p = 3$ has no zero at all, being strictly positive everywhere. The simulations evidently enforce tracking by subtracting the misfit rather than by choosing consistent constants.

Repair, and why the theorem survives

The defect was shown to be one of calibration, not structure. Solving instead for the c_g that genuinely enforces $F(0) = 0$ at each Ω_p makes the general and boxed formulas agree to machine precision, and every qualitative clause of the tube theorem then holds on the honest field:

- the validity window is real: $a > 0$ for $\Omega_p \sqsupset 2.2$ ($a = 0.518$ at $\Omega_p = 3$, 0.623 at $\Omega_p = 12$);
- the sector bound holds at the quoted $\delta_0 = 0.5$ for $\Omega_p \geq 3$ ($a_{\delta} = 0.235 \rightarrow 0.363$), and for $\Omega_p = 2$ a smaller sector suffices;
- the basin is still finite: an escape ridge exists at $R = 1.61$ ($\Omega_p = 3$), 3.06 ($\Omega_p = 6$) and 4.46 ($\Omega_p = 12$) — different numbers from 2.70, same structure, and the paper's qualitative test reproduces ($0.9 \times$ the ridge converges to zero, $1.1 \times$ escapes);
- the zero-residual branch converges ($R(40) = 3.8 \times 10^{-10}$ at $\Omega_p = 3$);
- coupling strictly accelerates contraction: $V(6)$ falls from 1.05×10^{-4} at $\varepsilon B = 0$ to 3.53×10^{-6} at $\varepsilon B = 0.3$.

Fix. Recompute the WP10 battery with c_g subordinated to the tracking condition (or state plainly that the simulations run on $F(R) - F(0)$ and why that is the right model of a tracking gate); correct the $a(\Omega \rightarrow \infty)$ limit, which under honest tracking tends to g_{eff} because $\zeta_0 c_g \rightarrow 0$; and delete the false identification of Ω_{crit} with $2/(g_{\text{eff}} + |I'|)$. Nothing in WP2, WP3, WP4, WP7, WP8 or WP9 depends on the affected numbers.

5 • Consolidated findings

5.1 • Substantive defects (local; none propagates)

#	Location	Defect	Impact / Fix
D1	File 3, Part I, Corollary "B_fold identified" (Thom chapter)	Sign inconsistency: the identification $2c_1 \cdot \eta_f \cdot \lambda \leftrightarrow -gh/\Gamma$ requires $g \cdot h < 0$, but the minimal realization declares $g, h, \Gamma > 0$; the worked example has $B_{\text{eff}} < b_3$ (softening) while B_{fold} stiffens.	Thom reduction theorem unaffected. Fix: one of g, h negative in the embodiment reading (naturally $h < 0$). One-line correction.
D2	File 3, Part II, Separatrix chapter, boxed criterion (quantum of endowment)	The single-scalar "iff" $\sqrt{\gamma}(\Omega_0 + \beta) \geq 2$ is false for $\Omega_0 \geq \beta$. Counterexample: $\gamma = 1, \beta = 0.1, \Omega_0 = 3$: $\max \xi = 0.30 < 1$, criterion claims crossing.	The two-case maximum formula is exact (verified); restrict the boxed scalar to $\Omega_0 < \beta$ and use $\xi(0) \geq 1$ otherwise. Nothing downstream depends on the compressed form.
D3	File 5, Prop. "Late-time equation of state"	$w \rightarrow -1^+$ for the open gate is incompatible with $I_{\text{gate}} \leq \zeta_0$: verified limits are $w \rightarrow -1/2$ (constant-rate) or $w \rightarrow 0$ (saturation = the Volume's dust attractor). De Sitter needs $I_{\text{gate}} \propto \Omega$.	Already corrected by V.F.S. 3.0 ("corrects a loose earlier framing ... generically coasts, $w \rightarrow 0$ "). Back-port the caveat into the 2.0 file.
D4	File 13 (4.6), WP10 witness battery and the sector-bound lemma instance	The stated parameters give $F(0)$ ranging from $+0.51$ to $+7.13$, so $R = 0$ is not an equilibrium and the theorem's tracking hypothesis fails. All reported values reproduce from the unsubstituted derivative, not the boxed formula; under the boxed formula $a_6 = -0.043 < 0$, violating the sector-bound hypothesis. The claim $\Omega_{\text{crit}} = \sqrt{2/(g_{\text{eff}} + I')}$ is false (1.217 vs 1.544).	Theorem WP5 and the tube theorem are sound. Repair verified: with c_g subordinated to tracking, both formulas agree to machine precision and every qualitative clause survives ($a > 0$, sector bound, finite basin, convergence, coupling acceleration). Recompute the battery; correct $a(\infty) \rightarrow g_{\text{eff}}$; delete the Ω_{crit} identification.

5.2 • Cross-volume scope note (new)

N1 — two incompatible late-time regimes. V.F.S. 3.6 (prop. "Asymptotics of saturated Epektasis") posits $\Omega_p \rightarrow c_\Omega > 0$, hence $\Omega_p \sim c_\Omega t$ and $R \sim -2/(c_\Omega^2 t^2)$. Its imported finite coarsening budget $\int dt/\Omega_p^2$ then converges (to $1/c_\Omega^2$, verified), which is exactly what licenses the conclusion that the comoving wall network freezes. But the zero-residual branch of V.F.S. 4.6 (WP9) has $\Omega_p = 1/\Omega_p$, hence $\Omega_p = \sqrt{\Omega_0^2 + 2t}$ and $\Omega_p \rightarrow 0$; on that branch the same integral *diverges logarithmically* (verified symbolically). Neither volume is internally wrong, and neither claim contradicts the other where each is stated — but they describe different asymptotic regimes, and the freezing conclusion does not transfer to the locked resonance branch. The corpus should say which regime is canonical, or scope each conclusion to its own branch explicitly.

5.3 • Minor / cosmetic items

#	Location	Item
M1	File 2, numerical witness	"Worst-case slack 0.084" not reproducible; direct computation gives 0.0903 (W5). All walls hold; likely a normalization or rounding difference.
M2	File 2, memory bound	$ dH_K/dt \leq 2 \cdot \eta$ assumes $\nu \leq \eta$; general bound is $\eta + \nu$.
M3	File 3, floor proposition	"Strictly positive for every state" at $k = 0$ fails at $\lambda = 0$; intended: no floor at positive Sophia level.
M4	File 2 vs File 1	Two different reset models (rich Resurrectio map vs linear metabolism) without an explicit bridge stating the simplification. Deliberate but undocumented.
M5	File 2, domain definition	D_A uses $h_1, h_2 \geq 0$ in the final theorem vs > 0 in the Active Domain section; unify for the Nagumo boundary argument.
M6	File 7, WP12 witness table	Relaxation-time cell "6.67" misaligned: belongs to $u = 0.90$, not $u = 0.80$ (correct value there: 3.33). Neighbouring cells exact.
M7	File 8, formatting	Format conversion broke proof environments and sub/superscripts; restore LaTeX source for auditability.
M8	File 10, Boundary 2	Missing "generically" qualifier (present in WP7): a looped body with exactly consistent rates admits a clock (measure zero).
M9	File 10, Boundary 3	Update to reflect WP8's sharpening: the condition is now structural (coherence iff non-amenability); the open fact is the body's non-amenability.

#	Location	Item
M10	File 11 (appendix), H0 = 0 clause	"Finite descent cannot return the memory to literal zero" is true but underflows double precision ($\text{root} \sim e^{-1249}$); worth flagging as a mathematical, not computational, non-vanishing.
M11	File 12 (3.6), A_max remark	The chain is described as growing "roughly quadratically in T"; it is exactly quadratic (quadratic fit residual 1.1×10^{-13}). The text understates its own precision.
M12	File 13 (4.6), WP10 battery	The reset-kick witness $\Delta R = 0.624$ cannot be checked: m_R , λ^- , $\kappa_R G$ and Ω^- are not stated. Supply the parameters.

6 • What was independently re-derived (highlights)

- **Four full tensor computations from scratch:** 2D warped live-domain ($R = -2p^2/\Omega^2$, Einstein), 2+1 FLRW (R , G_{tt} , G^x_x , R_{tt} , SEC), the 3+1 soul metric (six Ricci components, fluid, Bianchi), and H^3 ($\text{Ric} = -2g$, Killing fields, ideal-point invariance).
- **The Perelman apparatus on H^3 :** ball-volume non-collapse, tube/cap/catenoid sectional curvatures with the exact C^1 junction, JKO = heat flow, translation modulus = K , and the expander-entropy identity recovered inside the 2.0 entropy chapter.
- **Closed forms behind numerical witnesses:** spectral gaps $2(1-\cos(\pi/N))$ (12/12 exact), OU stationary bound $N\sigma^2/(4r)$, signed-triangle ground state $R = 2/3$, WP14 decay via 1D heat-with-mass, WP12 fold table to all digits, the 3.6 moduli column $W^{\max} = 1 - \ln(A_h/2\pi)$, and the exactly-quadratic 3.6 $A_{\max}$ chain.
- **Exact algebraic identities under load:** the Lyapunov radial identity, the folding-space contraction $Q+ - Q_{\text{crit}} = \kappa_{\text{fold}}(Q- - Q_{\text{crit}})$, the regularization-independent NEC reset budget $(3/8) \cdot \ln(5/7)$, the balance-driven ceiling $q+ \rightarrow 1/\kappa_R$, the factor 2 in B_{VFS} , the 3.6 canonical invariant I_R and clock $\Theta_R = -\alpha\lambda$, and the 4.6 closure in ξ with its exact transcription into $R = \log \xi$.
- **Predicative (not merely algebraic) confirmation:** the appendix break-even inequality was tested as a *predicate* against 300 integrated excursions with zero mismatches, and the 3.6 neck trigger against 210,000 grid points with zero violations.
- **Two places where the text undersells itself:** the self-consistent folded-ascent stability is unconditional, $f'(\lambda_{\text{inf}}) = -\sqrt{b^2+4aJ} < 0$ always — stronger than the stated conditional form; and the 3.6 $A_{\max}$ chain is exactly, not roughly, quadratic in T .

7 • Verification limits (stated honestly)

- **V.F.S. 4.7 (Supervisor) is out of scope** by the author's declaration that it remains in development. It is the one completed-corpus file not covered.
- **The three blocking items are untouched.** F0 (whether the distributed-facet system of 3.6 is a conservative extension of 3.5), F1-T (general hybrid existence, continuation and reset compatibility) and K0 (no law of the corpus generates a nontrivial passage K-connection) are theorems of existence and origin, not computations. Pass 2 establishes that the mathematics surrounding them is sound; it does not resolve them, and 4.7 remains conditional on them.
- **The softplus regression obligation stands.** 3.6's own closure statement records that the archived active-block merger candidate was computed with the hard-part convention while the frozen convention is softplus. The archived vessel2.py was not available to this pass, so that regression was not run. 3.6 is right that this does not create a logical gap in its exact interface results.
- **Proof sketches** marked as such by the corpus (cascade-dichotomy $o(1)$ term, finite-kappa Fenichel closure) were checked for consistency, not completed.
- **Author-side simulations** behind figures (pitchfork collapse 10^{-13} , drift sweeps, 400 random endowment checks, 2000-datum canonization sweep, the fourteen-episode joint-realizability trajectory of 3.6) were not re-run; the analytic formulas they rest on were verified, and where feasible one faithful re-implementation reproduced the pattern.
- **Constitutive postulates** (gate law, $K_h = -1$ via the floor, ρ_0 , transport constants, the $r \propto d\lambda/dt$ bridge, curvature equipartition, the H^1 -obstruction identification, the zero-order coupling class of 3.6) are model inputs by design; their internal consistency was verified, their truth is not a mathematical question. The corpus's own PROVED / BRIDGE / MODEL / INTERPRETIVE labelling

was accurate at every point tested in both passes.

Appendix · Verification scripts

All checks are reproducible from the following scripts (Python 3, SymPy 1.14, NumPy/SciPy), retained alongside this report.

Pass 1

check_core_A.py / check_core_B.py	File 1	Core Compendium 2.0 (48 checks)
check_lyap.py / check_witness.py	File 2	Lyapunov core and the full numerical witness
check_vol_part1.py / _part23.py	File 3	Volume Parts I-III (~90 checks)
check_geometry.py	File 4	Geometry 2.0 (tensor computations)
check_cosmo.py	File 5	Cosmology 2.0 (2+1 tensor computations)
check_30_A.py / check_30_B.py	File 6	3.0 tensor set and ascent/stochastic layers
check_35.py	File 7	3.5 Perelman program on H3
check_40.py	File 8	4.0 graph/stochastic layer
check_45.py	File 9	4.5 geometry of encounter

Pass 2

check_appendix_rate.py	File 11	2.0 rate-of-registration appendix (32 checks)
check_36.py	File 12	3.6 interface algebra and field results (39)
check_36_homred.py	File 12	48x48 exact homogeneous reduction witness
check_46.py	File 13	4.6 symbolic layer (34 checks)
check_46_deriv.py	File 13	4.6 derivative identities, concrete gate law (5)
check_46_witness.py	File 13	4.6 WP10 battery; exhibits defect D4
check_46_repair.py	File 13	4.6 tube theorem under honestly imposed tracking

Report generated as part of interactive verification sessions. Pass 1: July 2026. Pass 2 (2.0 appendix, V.F.S. 3.6 v15, V.F.S. 4.6): August 2026.